

Distinct running least common multiples

Affine tails and Erdős Problem #269

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ABSTRACT

We give a five-map proof that the reciprocals of the distinct running least common multiples of the 5-smooth integers have an irrational sum. The spacing of powers of 5 separates the tail images, so rationality would force a periodic jump sequence. We also derive a residue criterion for the sum counted with multiplicity, whose irrationality remains open here.

1 Two sums from running least common multiples

The 5-smooth integers are the positive integers whose prime factors belong to $\{2, 3, 5\}$. Their running least common multiples begin 1, 2, 6, 12, 60, 60, 120, 360, 360, For a finite set of primes P , let $\mathcal{S}_P$ consist of 1 and all positive integers whose prime factors belong to P , and put

$$L(x) = \text{lcm}\{u \in \mathcal{S}_P : u \leq x\}, \quad H_P(x) = \prod_{p \in P} p^{\lfloor \log_p x \rfloor} \quad (x \geq 1).$$

Every integer in this finite set divides $H_P(x)$. Conversely, the set contains the largest power of each $p \in P$ below or equal to x , so their product divides its least common multiple. Thus $L(x) = H_P(x)$. This value changes only at a positive prime power, where it is multiplied by that prime. Summing its distinct reciprocals gives

$$\mathcal{D}_P = 1 + \sum_{\substack{t=p^n \\ p \in P, n \geq 1}} \frac{1}{H_P(t)}.$$

Positive powers of distinct primes never coincide, and geometric growth of the successive running LCMs gives convergence. We use $\mathbb{N} = \{0, 1, 2, \dots\}$.

Theorem 1.1 (the sum over distinct running LCMs for $\{2, 3, 5\}$). *The number $\mathcal{D}_{\{2,3,5\}}$ is irrational.*

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If the sum were rational, its normalised tails would take only finitely many values. We show that each such value determines both the next block of jumps and the following tail. Repetition of a tail would then force the blocks to be eventually periodic, contrary to the irrational frequency of powers of 3. To recover a block from its tail, we use five affine maps whose images are separated by a bound obtained from the spacing of powers of 5.

Erdős asserted irrationality of $\mathcal{D}_P$ for every finite P with $|P| \geq 2$ in a letter dated 1 January 1973, without printing a proof [3, p. 335]. Thus the theorem above is a special case of that earlier assertion. We claim no priority for its conclusion. The [companion's general argument](#) uses an isolated interchange of jumps on a torus. That argument is retained there as an ordinary proof awaiting formalisation. The proof below is independent of that argument.

The question in Erdős Problem #269 concerns the sum with multiplicity,

$$\mathcal{R}_P = \sum_{u \in \mathcal{S}_P} \frac{1}{L(u)},$$

as recorded in [1, p. 65] and [2, p. 106]. For example, $L(5) = L(6) = 60$ for $P = \{2, 3, 5\}$, giving $2/60$ in $\mathcal{R}_P$ and $1/60$ in $\mathcal{D}_P$. The theorem treats the sum over distinct values; the problem asks about the sum with multiplicity. Our arguments leave the latter unresolved for finite P with $|P| \geq 3$. For $P = \{p\}$ both sums equal $p/(p-1)$.

Section 2 proves the theorem. We then retain the multiplicities and examine what changes: the tails satisfy an integer recurrence, but their values are unbounded. Sections 3 and 4 derive that recurrence and a residue criterion for the repeated sum. For two primes, Fan's factorisation [7] reduces both sums to a Hecke–Mahler value and yields transcendence (Section 5). Appendix A collects the elementary running LCM identities, and Appendix B explains why the three-prime kernel has no finite separated factorisation.

2 Recovering the prime-power jumps

For a rational number, long division has only finitely many possible remainders, and each remainder determines the next digit and remainder. The same two facts will prove useful here. Rationality bounds the number of tail values; the main work is to show that a tail determines the next block of prime-power jumps.

Proof of Theorem 1.1. Finitely many tails under rationality. Write $H = H_{\{2,3,5\}}$ in this proof and put $P_a = H(2^a)$. Every running LCM at or before 2^a divides P_a . We therefore normalise the tail after 2^a by

$$Y_a = P_a \sum_{\substack{t > 2^a \\ t=2^n, 3^n \text{ or } 5^n, n \geq 1}} \frac{1}{H(t)}. \quad (1)$$

Each subsequent jump multiplies the running LCM by at least 2, and a jump by 3 eventually occurs. Comparison with $\sum_{j \geq 1} 2^{-j}$ gives $0 < Y_a < 1$.

Suppose that $\mathcal{D}_{\{2,3,5\}} = N/K$, where N is an integer and $K \geq 1$ is an integer. Clearing the finite prefix gives

$$KY_a = P_a N - K \left(P_a + \sum_{\substack{t \leq 2^a \\ t=2^n, 3^n \text{ or } 5^n, n \geq 1}} \frac{P_a}{H(t)} \right) \in \mathbb{Z}.$$

Thus the Y_a lie in the finite set $(K^{-1}\mathbb{Z}) \cap (0, 1)$. It remains to show why equality of two tails propagates.

Recovering a block from its tail. Consider the jumps between 16 and 32, including the last endpoint. They occur at 25, 27, 32, multiplying the running LCM successively by 5, 3, 2. After normalisation at 16, their contribution is $1/5 + 1/15 + 1/30$. The remaining tail is normalised at a running LCM 30 times as large, so

$$Y_4 = \frac{1}{5} + \frac{1}{15} + \frac{1}{30} + \frac{Y_5}{30} = \frac{9 + Y_5}{30}.$$

In the block $(64, 128]$, the interior jumps occur in the opposite order, at 81, 125. Its constant term is $1/3 + 1/15 + 1/30 = 13/30$. The two blocks have the same multiplier 30, but different numerators, 9 and 13. To recover the order from a tail value, we must still separate the possible values of the block maps.

Every block $(2^a, 2^{a+1}]$ contains its terminal power of 2, at most one power of 3 and at most one power of 5. We record the interior jumps in their order by a word τ_a on the letters 3, 5, omitting the terminal 2. Splitting off the block gives

$$Y_a = G_{\tau_a}(Y_{a+1}), \quad G_{\tau}(y) = \frac{\mu_{\tau} + y}{b_{\tau}}, \quad \begin{array}{c|ccccc} \tau & \emptyset & 3 & 5 & 35 & 53 \\ \hline b_{\tau} & 2 & 6 & 10 & 30 & 30 \\ \mu_{\tau} & 1 & 3 & 3 & 13 & 9 \end{array} \quad (2)$$

Here b_{τ} is the factor by which the running LCM increases across the block, and μ_{τ}/b_{τ} is the block's normalised contribution.

To recover τ_a from Y_a , we need disjoint intervals containing the possible images. The bound $0 < Y_{a+1} < 1$ is insufficient, since $G_{53}([0, 1]) = [3/10, 1/3]$ and $G_5([0, 1]) = [3/10, 2/5]$ overlap. The table gives $Y_a \geq 3/10$. With this lower bound, separation of these two images requires an upper bound U satisfying

$$G_{53}(U) < G_5(3/10), \quad \frac{9 + U}{30} < \frac{33}{100}, \quad U < \frac{9}{10}.$$

It is therefore enough, for this pair of images, to improve the upper bound from 1 to a number below $9/10$.

A power 5^f belongs to block $\lfloor f \log_2 5 \rfloor$. These indices start at 2 and their successive differences are 2 or 3. Every three consecutive blocks therefore contain a power of 5. Given a , let $j \in \{a, a+1, a+2\}$ be the first such block. The maps for blocks containing 5 give $Y_j \leq 7/15$, since $Y_{j+1} < 1$ and $G_{35}(1) = 7/15$ bounds $G_5(1)$ and $G_{53}(1)$ as well. We now work *backwards* from j to a . Each intervening block has map $G_{\emptyset}$ or G_3 , and

$G_3(y) \leq G_\emptyset(y) = (1+y)/2$ for $y \geq 0$. At most two backward steps give the successive bounds $7/15, 11/15, 13/15$. Hence

$$Y_a \in I = \left[\frac{3}{10}, \frac{13}{15} \right], \quad G_\emptyset(G_\emptyset(7/15)) = \frac{13}{15} < \frac{9}{10}. \quad (3)$$

This bound holds for the tails of the prime-power sequence because every three consecutive blocks contain a power of 5. The interval I is not invariant under arbitrary compositions of the five maps: for example, $G_\emptyset(13/15) = 14/15$ lies outside I .

Figure 1 gives the exact images of I . They are pairwise disjoint, with successive gaps $1/900, 17/300, 79/900, 1/180$. Writing $L = 3/10$ and $U = 13/15$, the two narrow gaps are

$$G_5(L) - G_{53}(U) = \frac{3L - U}{30} = \frac{1}{900}, \quad G_\emptyset(L) - G_3(U) = \frac{3L - U}{6} = \frac{1}{180}.$$

Thus the same inequality $U < 3L = 9/10$ separates both pairs.

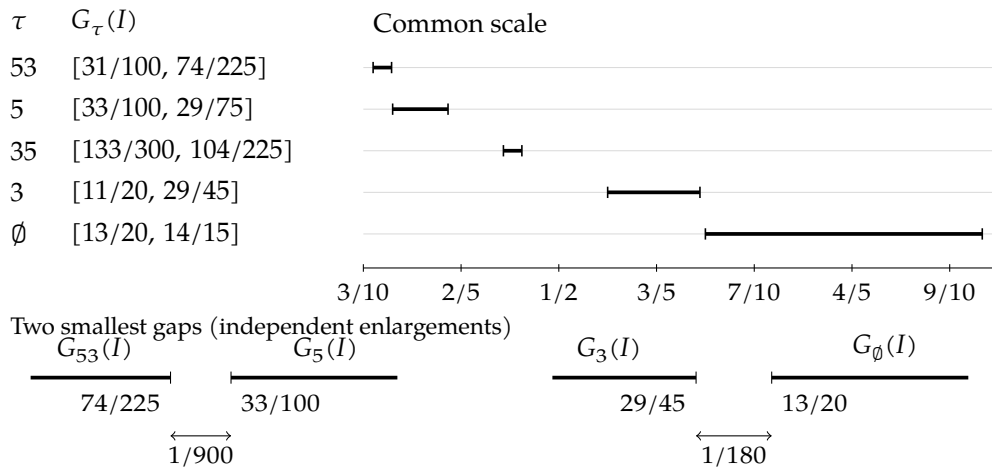


Figure 1: The five images of the arithmetic tail interval $I = [3/10, 13/15]$. Their separation determines the next block from an actual tail. The full image $G_\emptyset(I)$ extends beyond I ; the figure does not assert invariance under arbitrary words. Only the adjacent endpoints are shown in the two lower enlargements.

Since $Y_{a+1} \in I$, the actual tail Y_a belongs to exactly one of these five image intervals, identifying τ_a . For instance, $Y_a \in [31/100, 74/225]$ forces $\tau_a = 53$ and gives $Y_{a+1} = 30Y_a - 9$. In general we calculate $Y_{a+1} = b_{\tau_a} Y_a - \mu_{\tau_a}$, identify its image interval, and continue. Each recovered value is the next arithmetic tail, so the same bound I remains available at every step.

Repetition would give a rational frequency. The finitely many tail values obtained above include a repeated value $Y_u = Y_v$ with $u < v$. Since the image intervals are disjoint, this forces $\tau_u = \tau_v$. Applying the corresponding inverse map gives $Y_{u+1} = Y_{v+1}$. Induction yields $\tau_{u+j} = \tau_{v+j}$ for every $j \geq 0$. Thus the entire later block word has period $v - u$.

To exclude this possibility, let $\alpha = \log 2 / \log 3$. The number of powers of 3 in block a is $\lfloor (a+1)\alpha \rfloor - \lfloor a\alpha \rfloor$, and is determined by τ_a . If the block word has eventual period

$\ell \geq 1$, let c be the number of 3-powers in one period. For every sufficiently large fixed a and every $n \geq 1$ we then have

$$\lfloor (a + n\ell)\alpha \rfloor = \lfloor a\alpha \rfloor + nc.$$

Dividing by n and taking the limit gives $\ell\alpha = c$, contrary to unique factorisation, since it would imply $2^\ell = 3^c$. $\square$

3 The repeated three-prime sum

We return to the sum in which every smooth integer contributes. Put $P = \{2, 3, 5\}$, $S = \mathcal{R}_P$ and $H = H_P$ throughout this section. The running LCM still changes only at prime powers, but smooth integers between consecutive jumps also contribute to S .

For example, $[16, 32)$ contains the seven smooth integers 16, 18, 20, 24, 25, 27, 30. Grouping them by their running LCM gives

smooth integers	running LCM	contribution
16, 18, 20, 24	720	4/720
25	3600	1/3600
27, 30	10800	2/10800

Each of these three running LCMs contributes only once to $\mathcal{D}_P$. In the repeated sum their coefficients are 4, 1, 2, so the same jumps no longer give the same block contribution. Appendix A gives the finite grouping identity for general three-prime sets.

We use the half-open blocks $[2^a, 2^{a+1})$, so the tail starts at 2^a rather than just after it. Its finite prefix excludes the jump by 2 at that endpoint. Half the running LCM therefore clears that prefix: write

$$s_a = \sum_{\substack{x \in \mathcal{S}_P \\ 2^a \leq x < 2^{a+1}}} \frac{1}{H(x)}, \quad T_a = \sum_{j \geq 0} s_{a+j}, \quad h_a = \frac{H(2^a)}{2}, \quad X_a = h_a T_a.$$

Indeed, a running LCM strictly before 2^a has exponent of 2 at most $a - 1$ and divides h_a . Here $h_0 = 1/2$, while h_a is an integer for $a \geq 1$. This endpoint convention explains the factor $1/2$ and the difference from the tails in Section 2.

The change in normalisation is $b_a = H(2^{a+1}) / H(2^a)$. Splitting off s_a will give $X_{a+1} = b_a X_a - h_{a+1} s_a$, so we name the integer block contribution $m_a = h_{a+1} s_a$. Explicitly,

$$b_a = \frac{H(2^{a+1})}{H(2^a)}, \quad m_a = \sum_{\substack{x \in \mathcal{S}_P \\ 2^a \leq x < 2^{a+1}}} \frac{H(2^{a+1})}{2H(x)}. \quad (4)$$

For $[2, 4)$, the running LCMs at 2, 3 are 2, 6 and $h_2 = 6$, giving $m_1 = 6(1/2 + 1/6) = 4$ and $b_1 = 6$. For the block displayed above, $h_5 = 10800$ and

$$m_4 = 4 \cdot 15 + 3 + 2 \cdot 1 = 65, \quad b_4 = 30.$$

Thus m_a need not be smaller than b_a .

Lemma 3.1 (integer coefficients and four possible bases). *For every $a \geq 0$, m_a is a positive integer and $b_a \in \{2, 6, 10, 30\}$. The word “numerator” does not impose the positional-digit restriction $m_a < b_a$; that restriction need not hold.*

Proof. For $x < 2^{a+1}$ we have $2H(x) \mid H(2^{a+1})$, so every summand in m_a is integral. The interval contains 2^a , giving positivity. Between successive powers of 2 there is at most one power of 3 and at most one power of 5. Along with the final factor 2, these give $b_a \in \{2, 6, 10, 30\}$. $\square$

In the distinct-value proof, the ordered jumps in a block determine its numerator μ_τ . For the repeated sum, that word does not record how many smooth integers share each running LCM. We therefore retain the full block contribution m_a , not just the jumps.

Proposition 3.2 (the tail recurrence and a quadratic bound). *The series defining S, T_a converge. For every $a \geq 0$,*

$$X_{a+1} = b_a X_a - m_a, \quad 0 < X_a \leq \frac{8640}{343}(a+1)^2 < 90(a+1)^2.$$

For every integer $B \geq 1$, either BX_a is integral at some index (and hence at every later index), or $\text{dist}(BX_a, \mathbb{Z}) \geq 1/31$ at arbitrarily large indices.

Proof. We count at most $(a+1)^2$ smooth integers in $[2^a, 2^{a+1})$: each pair of exponents of 3, 5, both at most a , allows at most one exponent of 2. By (12), $s_a \leq 30(a+1)^2/8^a$. As $h_a \leq 8^a/2$ and $a+j+1 \leq (a+1)(j+1)$, we have

$$X_a \leq 15(a+1)^2 \sum_{j \geq 0} \frac{(j+1)^2}{8^j} = \frac{8640}{343}(a+1)^2.$$

This also proves convergence. Splitting off s_a gives the recurrence, since $h_{a+1} = b_a h_a$ and $m_a = h_{a+1} s_a$.

For the final assertion, suppose that every sufficiently late distance is strictly less than $1/31$, and write $BX_a = z_a + e_a$, where z_a is integral and $|e_a| < 1/31$. The recurrence makes $e_{a+1} - b_a e_a$ an integer of absolute value less than $(1+b_a)/31 \leq 1$. Thus $e_{a+1} = b_a e_a$. Since $b_a \geq 2$, boundedness forces every such error to vanish. Once integral, the values BX_a remain integral by the recurrence. $\square$

We can also write the repeated series as a Cantor series: $S/2 = \sum_{a \geq 0} m_a / (b_0 \cdots b_a)$. The criteria in [9, Theorem 2.1] and [10, Theorem 3.1] require a small-numerator hypothesis $m_a / (b_{a-1} b_a) \rightarrow 0$. Here the b_a are bounded and m_a grows quadratically. The [companion’s tail estimates](#) prove this lower bound as well as sharper upper bounds. Thus these criteria do not apply, and $X_a > m_a/30$ is unbounded. Rationality still places the tails in a fixed lattice, but no longer in a finite set as in Section 2. We shall use their quadratic bound to exclude possible integer tails instead.

Theorem 3.3 (rationality gives positive integer tails). *If $S = A/D$ in lowest terms, where $D = 2^u 3^v 5^w B$ and $\gcd(B, 30) = 1$, then for every $a \geq a_0 = u + 1 + 2v + 3w$,*

$$d_a = BX_a \in \mathbb{Z}_{>0}, \quad d_{a+1} = b_a d_a - B m_a, \quad d_a \leq 90B(a+1)^2.$$

Proof. For $a \geq 1$, clearing the finite prefix gives

$$X_a = h_a S - \sum_{\substack{x \in \mathcal{S}_p \\ x < 2^a}} \frac{h_a}{H(x)}, \quad \sum_{\substack{x \in \mathcal{S}_p \\ x < 2^a}} \frac{h_a}{H(x)} \in \mathbb{Z}. \quad (5)$$

If $a \geq u + 1 + 2v + 3w$, the exponent $a - 1$ of 2 in h_a is at least u , and $2^a \geq 3^v, 5^w$. Therefore $2^u 3^v 5^w \mid h_a$, and BX_a differs from $h_a A / (2^u 3^v 5^w)$ by an integer. Positivity, recurrence and the bound follow from Proposition 3.2. $\square$

Corollary 3.4 (the first index at which the denominator clears). *Under the same lowest-terms hypothesis, put $M = 2^u 3^v 5^w$ and let den denote the positive reduced denominator. For $a \geq 1$,*

$$\text{den}(BX_a) = \frac{M}{\gcd(M, h_a)}.$$

Consequently BX_a is integral exactly when $2^a \geq \max(2^{u+1}, 3^v, 5^w)$. The first such a can be found by integer comparisons, without logarithmic rounding.

Proof. Equation (5) shows that BX_a and $h_a A / M$ have the same reduced denominator. Since $\gcd(A, M) = 1$, this denominator is $M / \gcd(M, h_a)$. Divisibility by each of $2^u, 3^v, 5^w$ gives the three stated inequalities. $\square$

For a hypothetical reduced denominator $2^3 3^2 5 \cdot 7$, the first integral $7X_a$ would occur at $a = 4$. The sufficient starting index in the theorem is 11. The [companion's denominator calculation](#) also determines the reduced denominator of X_a itself.

4 A residue criterion

If S is rational, the values $d_a = BX_a$ are eventually positive integers bounded by $90B(a + 1)^2$. Iterating the recurrence places each later value in a particular residue class. We shall exclude integrality when the least positive integer in this class exceeds the bound.

For two steps the recurrence reads $X_{\ell+2} = b_{\ell+1}b_\ell X_\ell - (b_{\ell+1}m_\ell + m_{\ell+1})$. To keep track of the multiplicative and additive terms over h steps, write

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \quad W_{\ell,h+1} = b_{\ell+h}W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h}F_{\ell,h} + m_{\ell+h}.$$

Induction yields

$$X_{\ell+h} = W_{\ell,h}X_\ell - F_{\ell,h}. \quad (6)$$

The quotient $F_{\ell,h}/W_{\ell,h}$ is the contribution of these h blocks to the starting tail:

$$\frac{F_{\ell,h}}{W_{\ell,h}} = h_\ell \sum_{a=\ell}^{\ell+h-1} s_a.$$

The remainder is $X_{\ell+h}/W_{\ell,h}$. Thus (6) splits X_ℓ into a finite sum and its tail, with both measured in the starting normalisation. Throughout this section let

$$K(B, a) = 90B(a + 1)^2. \quad (7)$$

The bound $0 < BX_{\ell+h} \leq K(B, \ell + h)$ and (6) enclose the starting tail in the interval

$$\frac{BF_{\ell,h}}{W_{\ell,h}} < BX_{\ell} \leq \frac{BF_{\ell,h} + K(B, \ell + h)}{W_{\ell,h}}.$$

Its width is $K(B, \ell + h)/W_{\ell,h}$, but a width below 1 does not by itself exclude an integer. We must compare its right endpoint with the first integer strictly above its left endpoint. The residue test makes this comparison at the other end of the segment: an integral starting tail would make $BX_{\ell+h}$ a positive integer congruent to $-BF_{\ell,h}$ modulo $W_{\ell,h}$.

For integers $W \geq 1, t$, define $\text{lpr}_W(t) = 1 + ((t - 1) \bmod W)$, with the remainder in $\{0, \dots, W - 1\}$. This is the least positive integer in the residue class of t . In particular $\text{lpr}_W(0) = W$, rather than zero: the left endpoint of the interval is excluded because the terminal tail is positive.

Lemma 4.1 (least positive residues). *If d is a positive integer with $d \leq K$ and $d \equiv -BF \pmod{W}$, where $W \geq 1$, then $\text{lpr}_W(-BF) \leq K$.*

Proof. The least positive representative is at most every positive representative of its class, and in particular at most d . $\square$

For $\ell = 1, h = 6$, direct iteration gives $W_{1,6} = 648000$ and $F_{1,6} = 524431$. Since $K(1, 7) = 5760$, the interval is

$$\frac{524431}{648000} < X_1 \leq \frac{530191}{648000} < 1.$$

It contains no integer. Equivalently, $\text{lpr}_{W_{1,6}}(-F_{1,6}) = 123569 > 5760$ excludes an integral X_1 . A rational denominator might clear only later. The criterion must therefore test every denominator part coprime to 30 beyond any prescribed clearing index, rather than just at one early start.

Theorem 4.2 (a residue criterion for irrationality). *The number S is irrational if and only if*

$$\begin{aligned} &\text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1, \\ &\text{there are } \ell \geq a_0, h \geq 1 \text{ such that } \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > K(B, \ell + h). \end{aligned} \quad (8)$$

Proof. Suppose that (8) holds and S is rational. By Theorem 3.3, $d_a = BX_a$ is a positive integer for every sufficiently large a . Choose a segment beyond that onset. Equation (6) gives $d_{\ell+h} \equiv -BF_{\ell,h} \pmod{W_{\ell,h}}$, contrary to Lemma 4.1 and (8).

Conversely, suppose that S is irrational and fix $B, \ell \geq 1$. Equation (5) makes BX_{ℓ} nonintegral. The distance upwards to the next integer, $\delta = \lceil BX_{\ell} \rceil - BX_{\ell} \in (0, 1)$, is now fixed as the segment length h grows. We have $W_{\ell,h} \geq 2^h$, whereas $X_{\ell+h}$ has only quadratic growth. Consequently $\delta + BX_{\ell+h}/W_{\ell,h}$ lies in $(0, 1)$ for sufficiently large h . This ensures that the following representative lies strictly between 0 and the modulus, so it is the least positive one:

$$\text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \lceil BX_{\ell} \rceil W_{\ell,h} - BF_{\ell,h} = \delta W_{\ell,h} + BX_{\ell+h}.$$

The positive constant δ multiplies an exponentially growing $W_{\ell,h}$, so the first term eventually exceeds $K(B, \ell + h)$. This proves the inequality from every fixed start, and hence after every prescribed onset a_0 . No bound uniform in B or ℓ is needed for this direction. $\square$

As h grows, the interval above narrows towards the fixed value BX_ℓ . If that value is nonintegral, a sufficiently narrow interval contains no integer. Failure to exclude an integer at a finite length proves neither integrality nor failure at every longer length. The converse gives no estimate for the gap δ ; to use the forward implication, we must prove (8) for the actual coefficients without assuming that nonintegrality.

The upper-bound property of K is essential: choosing the zero function would make every residue inequality automatic. The [companion's residue analysis](#) treats smaller bounds and identifies when more general bounds give the same criterion. Finite searches cover only finitely many multipliers and starts. The repeated three-prime question can therefore be written in three equivalent forms.

Problem 4.3 (the repeated three-prime target). Prove that $S = \mathcal{R}_{\{2,3,5\}}$ is irrational.

Problem 4.4 (no reduced tail is an integer). Prove that for every $a \geq 1$ and every integer $B \geq 1$ coprime to 30,

$$BX_a \notin \mathbb{Z}. \quad (9)$$

Problem 4.5 (residue inequalities after every starting index). Prove (8).

An integral BX_a makes S rational by (5); Theorem 3.3 proves the converse implication. Theorem 4.2 gives the third form. All three remain unproved for S . The [companion's discussion of the repeated sum](#) records the finite calculations and the hypotheses missing from proposed finite-difference and function-theoretic arguments.

5 Two-prime transcendence

For two primes we can retain the multiplicities and still evaluate the sum through one Hecke–Mahler value. Fan gave the factorisation of the repeated sum and its transcendence consequence in his post of 26 June 2026 [7]. The following calculation also expresses the sum over distinct running LCMs in that value.

Theorem 5.1 (both two-prime sums). *Let $p < q$ be distinct primes. Put $\theta = \log p / \log q$ and $A = \sum_{n \geq 0} p^{-n} q^{-\lfloor n\theta \rfloor}$. Let $\mathcal{R}_{p,q}$ sum the reciprocal running LCM at every positive $\{p, q\}$ -smooth integer, and let $\mathcal{D}_{p,q}$ count each distinct running LCM once. Then*

$$\mathcal{D}_{p,q} = \frac{(q-p)A + p}{q-1}, \quad \mathcal{R}_{p,q} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \quad (10)$$

Both numbers are transcendental.

The Lean proof assumes the transcendence theorem of Bugeaud and Laurent; the two identities in (10) are proved in Lean without it.

Proof. Write $x = 1/p$, $y = 1/q$, $m_n = \lfloor n\theta \rfloor$ and $\delta_n = m_{n+1} - m_n$. Unique factorisation makes θ irrational, and $0 < \theta < 1$ gives $\delta_n \in \{0, 1\}$. The initial value and the positive powers of p contribute $A = \sum_{n \geq 0} x^n y^{m_n}$. There is one q -power strictly between p^n and p^{n+1} exactly when $\delta_n = 1$, so the positive powers of q contribute

$$B = \sum_{n \geq 0} \delta_n x^n y^{m_n+1}, \quad \mathcal{D}_{p,q} = A + B.$$

Since $0 < x, y < 1$, these series converge absolutely. The identity $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n} (y - 1)$ and an index shift in A give $A - 1 - xA = x(y - 1)B/y$. Therefore

$$B = \frac{p - (p - 1)A}{q - 1}.$$

For the repeated sum, the running LCM at $p^i q^j$ is $p^{i+\lfloor j/\theta \rfloor} q^{j+m_i}$. Its reciprocal is the product of $x^i y^{m_i}$, which depends only on i , and $y^j x^{\lfloor j/\theta \rfloor}$, which depends only on j . Absolute convergence therefore gives

$$\mathcal{R}_{p,q} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor} = A(1 + B).$$

For the last equality, each $j \geq 1$ corresponds to $n = \lfloor j/\theta \rfloor$ with $m_n = j - 1$ and $\delta_n = 1$. Substitution proves (10).

Both sums are now nonconstant polynomials in the single value A , with rational coefficients. It therefore remains to prove that A is transcendental. For this we use the Hecke–Mahler series $F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{m_n} x^n y^k$. Geometric summation gives

$$A = \frac{1}{1-x} - \frac{1-y}{y} F_\theta(x, y). \quad (11)$$

Bugeaud and Laurent’s theorem [5, Theorem 1.1] applies with intercept $\rho = 0$, $\beta = x$ and $\alpha = y$: the slope θ is irrational and lies in $(0, 1)$, x and y are nonzero algebraic numbers, $|x| < 1$ and $|xy^\theta| = p^{-2} < 1$. Thus $F_\theta(x, y)$ is transcendental; this case $\rho = 0$ is due to Loxton and van der Poorten [6, Theorem 8, p. 40]. Thus A is transcendental. The displayed affine and quadratic polynomials are nonconstant, so an algebraic value of either would force A to be algebraic. $\square$

The [companion’s two-prime calculation](#) records an ordinary extension to the monoid generated by coprime integers $1 < p < q$, such as 4, 9. That monoid need not contain all integers supported on the prime divisors of pq ; multiplicatively dependent generators such as 4, 8 require a different calculation.

For larger prime sets, the sums over pure powers $E_p = \sum_{\alpha \geq 0} L(p^\alpha)^{-1}$ satisfy $\mathcal{D}_P = \sum_{p \in P} E_p - (|P| - 1)$. The [companion’s prime-power subseries argument](#) treats their irrationality as a remark with an ordinary proof awaiting formalisation. The source record contains no separate priority search for this claim. Irrationality of the individual subseries would not imply irrationality of their sum. The repeated series also contains mixed products of primes, whose multiplicities change the tail estimates.

A Identities for the running LCM and multiplicities

Let p, q, r be pairwise distinct primes. Write $H = H_{\{p,q,r\}}$ and $K(i, j, k) = H(p^i q^j r^k)^{-1}$ throughout this appendix and the next. The following identities underlie the grouping of the repeated series in Section 3.

Proposition A.1 (the running least common multiple). *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then the running least common multiple is given by $L(x) = H(x)$.*

Proof. We have $u \mid H(x)$ for every supported $u \leq x$. For the reverse divisibility, take the maximal powers of p, q, r below x : they belong to the prefix and are pairwise coprime, so their product divides its LCM. $\square$

For each prime, $x/p < p^{\lfloor \log_p x \rfloor} \leq x$. Multiplying gives

$$x^3 / (pqr) < H(x) \leq x^3. \quad (12)$$

Proposition A.2 (cells and jumps). *The running LCM is constant when the three integer logarithms are constant. A jump in exactly one logarithm multiplies it by the corresponding prime. The first n positive powers of each prime, together with 1, form $3n + 1$ distinct points.*

Proof. The running LCM is a product of the three maximal pure powers, which proves the constancy and jump assertions. Unique factorisation separates the $3n$ positive pure powers from one another and from 1. $\square$

Proposition A.3 (grouping terms with the same running LCM). *For a finite exponent box $\mathcal{B}$, set $F(H) = \{(i, j, k) \in \mathcal{B} : H(p^i q^j r^k) = H\}$. Then*

$$\sum_{(i,j,k) \in \mathcal{B}} K(i, j, k) = \sum_H \frac{\#F(H)}{H}. \quad (13)$$

Proof. Partition the finite exponent box by the value of the running LCM. The terms in the part $F(H)$ all equal $1/H$. $\square$

B The three-prime kernel

We ask whether the two-prime factorisation can be replaced, for three primes, by a finite sum of products separating one exponent from the other two. The kernel $K(i, j, k) = H(p^i q^j r^k)^{-1}$ rules this out. Diagonal rescaling reduces the question to a matrix with entries 1 and $1/r$; separate choices of row and column indices then give nonzero minors of every order.

Example B.1. For $(p, q, r) = (2, 3, 5)$, the leading two-by-two determinant is $-1/15$: its four entries are $1, 1/6, 1/2, 1/60$, so it equals $1/60 - 1/12$.

This nonzero two-by-two minor rules out rank one. More generally, a sum of d separated products would give every matrix layer rank at most d . To exclude every finite d , we therefore seek nonzero minors of arbitrarily large order. Their rows and columns may depend on the order; the same choices will work in every layer k .

Theorem B.2 (no finite separation of the kernel). *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 0$ there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(K(I(a), J(b), k))_{0 \leq a, b < n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell(i)$ and $G_\ell(j, k)$, $0 \leq \ell < d$, satisfying

$$K(i, j, k) = \sum_{\ell < d} f_\ell(i) G_\ell(j, k) \quad \text{for all } i, j, k.$$

Proof. For fixed k , the powers of p and q split into a row factor and a column factor, as in the two-prime case. The power of r has one remaining interaction between the indices. Put $c = r^{-1}$, $x_i = \{i \log_r p\}$ and $y_j = \{j \log_r q\}$, where braces denote fractional parts. Then

$$\lfloor i \log_r p + j \log_r q \rfloor = \lfloor i \log_r p \rfloor + \lfloor j \log_r q \rfloor + \mathbf{1}_{\{x_i + y_j \geq 1\}}.$$

The last term is the carry when the two fractional parts are added. Putting all the other prime powers into row and column factors gives

$$K(i, j, k) = U_i(k)^{-1} C_{ij} V_j(k)^{-1}, \quad C_{ij} = c^{\mathbf{1}_{\{x_i + y_j \geq 1\}}},$$

where

$$U_i(k) = p^i q^{\lfloor \log_q(p^i r^k) \rfloor} r^{k + \lfloor \log_r p^i \rfloor},$$

$$V_j(k) = q^j p^{\lfloor \log_p(q^j r^k) \rfloor} r^{\lfloor \log_r q^j \rfloor}.$$

These positive factors preserve nonvanishing of minors. In column j , the entry changes from 1 to c when x_i reaches $1 - y_j$. Distinct primes make $\log_r p$ and $\log_r q$ irrational, so each fractional-part orbit is dense in $[0, 1]$. We choose row and column indices independently, using only these two one-dimensional density statements. For $n \geq 1$, choose distinct rows with $0 < x_{I(0)} < \dots < x_{I(n-1)} < 1$. We choose each column so that its entries change from 1 to c at a different selected row. Writing $s_b = 1 - y_{J(b)}$, density of (y_j) lets us choose

$$s_0 \in (0, x_{I(0)}), \quad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \leq b < n).$$

The intervals for the s_b are disjoint, so the column indices are distinct. The strict inequalities also avoid the threshold itself, and give $x_{I(a)} + y_{J(b)} \geq 1$ exactly when $b \leq a$. Thus multiplying row a by $U_{I(a)}(k)$ and column b by $V_{J(b)}(k)$ reduces the selected kernel minor to

$$T_n(c) = \begin{pmatrix} c & 1 & \dots & 1 \\ c & c & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ c & c & \dots & c \end{pmatrix}, \quad \det T_n(c) = c(c-1)^{n-1}.$$

Indeed, subtract each preceding row from the next, working upwards from the last row. Each row difference has just one nonzero entry, $c - 1$, in its own column. The resulting triangular matrix has diagonal entries $c, c - 1, \dots, c - 1$. The original determinant is therefore

$$c(c-1)^{n-1} \prod_{a < n} U_{I(a)}(k)^{-1} \prod_{b < n} V_{J(b)}(k)^{-1} \neq 0.$$

The indices were chosen from C , independently of k . For $n = 0$ the empty determinant equals 1.

For the final assertion, suppose a separation with d summands existed and fix any k . On the rows $I(0), \dots, I(d)$ and columns $J(0), \dots, J(d)$ its matrix would be the product of the $(d+1) \times d$ matrix $(f_\ell(I(a)))_{a,\ell}$ and the $d \times (d+1)$ matrix $(G_\ell(J(b), k))_{\ell,b}$. Its rank would be at most d , so its determinant would vanish, contrary to the minor just constructed. $\square$

No continuity or boundedness is assumed for the separated factors.

Corollary B.3 (the same minors modulo integers coprime to 30). *For $(p, q, r) = (2, 3, 5)$ and every $n \geq 1$, there are injective maps $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$, chosen independently of B and k , such that for every $B \geq 2$ coprime to 30 and every $k \geq 0$, the selected $n \times n$ kernel matrix has unit determinant over $\mathbb{Z}/B\mathbb{Z}$, with each reciprocal prime power interpreted by its modular inverse.*

Proof. Choose the maps from Theorem B.2. Every row and column factor is a unit modulo B . The normalised determinant is $5^{-1}(-4/5)^{n-1}$, also a unit. $\square$

The modulus may be composite, for example 49 or 77. Coprimality with 30 makes both the reciprocal entries and the determinant units: the latter introduces only a power of 4 in its numerator.

Leading minors. The leading 4×4 block at $\{2, 3, 5\}$ is singular:

$$K(3, j, 0) = \frac{1}{120} K(0, j, 0) \quad (0 \leq j < 4).$$

These equalities follow by evaluating the running LCM at 3^j and $8 \cdot 3^j$. But at $j = 4$, $K(3, 4, 0) - K(0, 4, 0)/120 = -1/19440000$. Thus the leading minors do not supply the arbitrary-order theorem.

Proposition B.4 (rank of a matrix of threshold columns). *Let $m \geq 1$ and let c lie in a field with $c \neq 0, 1$. For $0 \leq k \leq m$ write v_k for the length- m column with a 1 in each of the first k coordinates and c thereafter. A matrix whose distinct columns are v_k for k in a nonempty set E has rank $|E| - \mathbf{1}_{\{0, m\} \subseteq E}$.*

The correction accounts for the two constant columns: $v_0 = cv_m$. There is no other dependence among distinct threshold columns.

Proof. List $E = \{k_1 < \dots < k_t\}$. The $t-1$ differences $v_{k_{j+1}} - v_{k_j} = (1-c)\mathbf{1}_{\{k_j, \dots, k_{j+1}-1\}}$ have disjoint nonempty supports, hence are linearly independent. If $k_1 > 0$ or $k_t < m$, their union misses a coordinate where v_{k_1} is nonzero, so the rank is t . If $k_1 = 0$ and $k_t = m$, then $v_0 = c\mathbf{1}$ lies in the span of the differences (their sum is $(1-c)\mathbf{1}$), so the rank is $t-1$. $\square$

The restrictions $c \neq 0, 1$ exclude a zero column or identical columns; they hold for $c = 1/r$ over $\mathbb{Q}$. In one fixed layer k , ordering the sampled row phases puts every column in the stated form. The formula therefore determines the rank of each nonempty rectangular sample. For fixed row and column indices this rank is independent of k . The [companion's kernel analysis](#) gives the exact rational comparisons.

Approximation. Truncating the original kernel to $i < N$ gives a sum of N separated terms. The bound $H(x) > x^3/(pqr)$ makes the truncation error tend to zero both uniformly and in $\ell^1(\mathbb{N}^3)$. The rescaled matrix in the rank proof has uniform distance $(1-1/r)/2$ from the matrices of finite rank. Rescaling preserves exact rank but removes the decay responsible for the first approximation. The [companion's approximation theorem](#) proves both assertions for their stated norms. Neither the rank theorem nor the approximation bound implies irrationality of the scalar sum.

C Verification, reproducibility and sources

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded. The five-map proof is in [the distinct-value development](#) and [its block-series criterion](#). Its statement has no recorded Comparator comparison. The evidence record fixes the source versions; the named input for two-prime transcendence is stated beside that theorem.

The [programs for sums over distinct values](#) calculate the interval endpoints and finite-state examples using exact arithmetic; the proof in Section 2 derives its constants directly.

The [companion’s account of this proof](#) also compares one-jump and dyadic bounds. Its finite-state extension uses intervals depending on earlier blocks. The saved exact computations satisfy that criterion for every set of three primes at most 31 and for 292 of the 330 four-prime sets in that range. The general criterion and those computations are recorded as remarks awaiting formalisation. For $\{2, 3, 5, 7\}$ two admissible automaton words give the same affine map at every subsequent refinement, preventing that automaton from recovering its first block. This identity does not establish a collision between two words realised by the arithmetic sequence.

The [companion’s source inventory](#) also covers the ordinary general and subseries arguments. Its historical note records the 26 September 2026 AI check and the corrections made from it; independent human mathematical review has not been reported.

Further comparisons

The classical criteria of Erdős–Straus [9], Hančl–Tijdeman [10] and Diananda–Oppenheim [8] are compared with the bounded-base series in the companion. It also compares the bounded-ratio argument with Erdős–Taylor [11, Theorem 1, p. 600] and Fan [12, Lemma 3.1, p. 7], and the restrictions on integer recurrences with the polynomial Cantor-series results of Hančl–Tijdeman [13, Theorems 2.2, 3.1 and 4.2]. Koutsoukou–Argyaki and Li’s Isabelle development [19] formalises classical irrationality criteria. It does not treat the present repeated series.

For transcendence, the relevant comparisons are the finite-difference conditions of Luca–Ouaknine–Worrell [14], the echoing conditions in [15], the fixed-base complexity theorem of Adamczewski–Bugeaud [20], and multivariate Mahler theory [21]. Quadratic growth of m_a supplies none of the missing cancellation or recurrence hypotheses. The companion exhibits nonzero boundary terms in a proposed third-difference calculation and distinguishes variable-base digits from fixed-base expansions. Work on fixed-prime semigroups [16, 17] and other Ahmes-series problems [18] provides context; no estimate from those papers enters our dyadic counting argument.

Statements and declarations

Authorship and availability. AI agents carried out most of the research and drafting. The companion contains the longer proofs and exact computations, together with the hypotheses and limitations of the unsuccessful approaches.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator pending	Lemma 3.1	Lean · Comparator
Proposition 3.2	Lean · Comparator	Theorem 3.3	Lean · Comparator
Corollary 3.4	Lean · Comparator	Lemma 4.1	Lean · Comparator
Theorem 4.2	Lean · Comparator	Theorem 5.1	Lean [†]

Statement	Formal proof	Statement	Formal proof
Proposition A.1	Lean · Comparator	Proposition A.2	Lean · Comparator
Proposition A.3	Lean · Comparator	Theorem B.2	Lean · Comparator
Corollary B.3	Lean · Comparator	Proposition B.4	Lean · Comparator